1-DoF Grasp-and-Twist Underactuated Gripper with Differential Tendon Mechanism

ME5250 Robot Mechanics and Control — Project 1 Report

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Abstract

A **1-DOF underactuated gripper** performs finger closure and in-hand wrist rotation from a *single* position-controlled actuator—no sensors, no switching logic. Two parallelogram four-bar fingers ($L_0 = L_2 = 12\,\text{mm}$, $L_1 = L_3 = 40\,\text{mm}$) with spring-loaded distal phalanges and a passive wrist joint are coupled by a compound tendon differential: $\ell_{\text{master}} = 1.0\,\theta_R + 1.0\,\theta_L + 0.25\,\varphi$. Stiffness contrast between the free fingers and wrist spring ($k_w = 5.0\,\text{N·m/rad}$) produces an *approximately* sequential mode switch: fingers close first, then the wrist rotates once actuator force exceeds the spring threshold. In practice, object contact stalls the fingers before the joint limit, yielding a compliant blending zone rather than a sharp transition. The gripper has **4 DOFs and 1 actuator**; MuJoCo 3.3.3 simulations validate differential behavior across cylinder, sphere, and box geometries.

Video URL: youtu.be/fcBXoY75yp8

GitHub Repository: keivalya/me5250-project-1

1. Introduction and Literature Survey

Industrial manipulation often requires a gripper to both *secure* and *reorient* an object—picking a cap from a bin, then rotating it upright for capping. The standard fix is a dedicated wrist actuator, but each added motor brings cost, weight, and control overhead [1]. Under-actuation offers a cleaner path: encode the sequencing into the mechanism itself so one actuator produces multiple outputs through passive force-threshold switching.

Birglen, Laliberté, and Gosselin [1] formalized this with the transmission-matrix framework, showing how tendon and linkage networks distribute torque across passive joints. Dollar and Howe [3] validated the single-actuator paradigm with the SDM Hand, which conforms to irregular geometry through contact-induced deflection. Ma and Dollar [4] made the approach more accessible with the Yale OpenHand Project, an open-source library of 3D-printable designs. Commercially, the Robotiq 2F-85 [9] uses the same four-bar parallelogram topology adopted here; its MuJoCo Menagerie model served as a direct reference. Yoon and Choi [5] provide another example of underactuated finger design for robotic hands, reinforcing the broader literature trend toward mechanically embedded adaptation.

Two works most directly motivate this design. Nishimura *et al.* [6] achieved *infinite* in-hand rotation from a single actuator via a differential *gear*, mode switching triggered by force threshold alone. The SPINE Gripper [7] shows tendon geometry alone can produce out-of-plane reorientation. An earlier companion paper [8] confirmed the sensor-less switching principle holds across mechanism types.

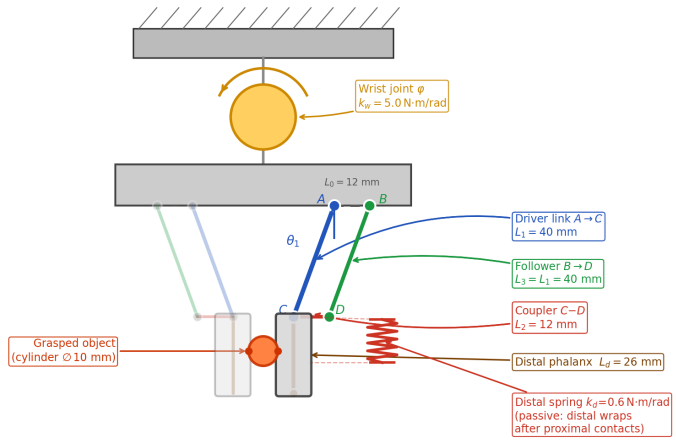

Figure 1: Parallelogram four-bar linkage (right finger in full colour; left mirrored and faded). A–B: ground link $L_0 = 12\,\text{mm}$, fixed to base (dashed). A–C: driver $L_1 = 40\,\text{mm}$ (**blue**). B–D: follower $L_3 = L_1$ (**green**). C–D: coupler $L_2 = 12\,\text{mm}$ (**red**). Gray rectangles: distal phalanges ($L_d = 26\,\text{mm}$). Red zigzag: passive distal spring $k_d = 0.6\,\text{N·m/rad}$ — after the proximal pad contacts the cylinder, the distal phalanx rotates against k_d to conform to the surface. Amber circle: wrist joint φ, passive spring $k_w = 5.0\,\text{N·m/rad}$ (Phase 2 rotation). Driver angle θ_1 at pivot A.

Design contribution. This gripper applies the same mode-switching logic through a compound `<fixed>` tendon in MuJoCo, replacing the gear differential with a purely tendon-based implementation. Spring-loaded distal phalanges add a third passive DOF for fingertip shape adaptation. The MJCF file is the functional CAD model—bodies, joints, dimensions, and constraints defined once, exercised in simulation. The result is a mechanism with three passive DOFs serving *qualitatively different* functions (wrist rotation, right- and left-finger conformation) from one position command, with no sensors.

2. Mechanism Design

2.1 Kinematic Tree

The gripper is a serial–parallel hybrid. The top-level tree is:

`mount` → `wrist_body` → `base_mount` → `fingers`

The mount is a rigid ceiling fixture (cylinder, radius 22 mm). `wrist_body` carries all downstream links and rotates about $\hat{z}$ on the wrist joint. `base_mount` ($80 \times 30 \times 12$ mm box) is the structural backbone from which both fingers hang.

2.2 Parallelogram Four-Bar Fingers

Each finger is a planar parallelogram four-bar linkage (Fig. 1) with equal opposite links:

$$L_0 = L_2 = 12\,\text{mm}, \quad L_1 = L_3 = 40\,\text{mm}. \tag{1}$$

The driver pivot (A) sits at ± 24 mm from the centreline and the follower pivot (B) at ± 36 mm, giving $L_0 = 12$ mm. The parallelogram condition keeps the coupler at *constant orientation* throughout motion—the distal pad stays parallel to the base at any driver angle. The driver joint is tendon-actuated over $\theta_1 \in [0,\ 0.8]$ rad ($\approx 46°$), with the closed loop enforced by MuJoCo constraints.

2.3 Spring-Loaded Distal Phalanges

A distal body ($L_d = 26$ mm) attaches at coupler joint C, *outside* the four-bar loop, on a passive torsion spring ($k_d = 0.6$ N·m/rad, $b_d = 0.25$ N·m·s/rad, $\delta \in [-0.5,\ 1.2]$ rad). At rest $\delta = 0$; under contact, the object deflects the joint against the spring, wrapping the fingertip to the surface without adding actuator. This accounts for two of the three passive DoFs.

2.4 Wrist Joint

The wrist joint is revolute about the vertical axis $\hat{z}$, loaded by:

$$k_w = 5\,\text{N}\cdot\text{m/rad}, \quad b_w = 1.5\,\text{N}\cdot\text{m}\cdot\text{s/rad}, \quad \varphi \in [-2\pi,\ 2\pi]. \tag{2}$$

The stiffness k_w is chosen to be large relative to the zero-load finger torque, strongly biasing the mechanism toward finger closure in the initial part of the motion. However, in contact, the wrist can still deflect slightly before a fully sharp phase boundary is reached.

2.5 Compound Tendon Differential

A single MuJoCo tendon couples all three movable subsystems through one scalar input:

$$\ell_{\text{master}} = 1.0 \cdot \theta_R + 1.0 \cdot \theta_L + 0.25 \cdot \varphi. \tag{3}$$

One position-controlled actuator ($k_p = 80$, force range ± 150 N, $\ell_{\text{ctrl}} \in [0,\ 2.5]$) drives this tendon.

The differential works by contrast to stiffness. Fingers do not have intrinsic spring resistance, so tendon shortening initially goes *predominantly* into $\theta_{R,L}$, while the wrist spring ($k_w = 5.0$ N·m/rad) biases the mechanism against rotation. In the ideal no-contact limit this yields a clean two-stage response; in contact, a small amount of wrist compliance can appear before a perfectly sharp boundary is reached. Once the fingers stall, either on object contact or at the joint limit, continued actuator force is redirected primarily into the wrist. The transition is governed entirely by torque balance: *no control-law change, no sensor, no external trigger*.

3. Mobility and DOF Analysis

Table 1 gives the complete Grübler analysis. For a planar mechanism, $M = 3(N-1) - 2J_1$. Per finger: $N = 4$ links, $J_1 = 4$ revolute joints, so $M = 1$ DOF. The symmetric tendon constraint ($\theta_R \equiv \theta_L$) reduces the two-finger count to 1 actuated DOF.

Table 1: Grübler mobility analysis of the gripper.

Component	**DOF**	**Type**
Structural (four-bar per finger):		
N=4 links, J_1=4 revolutes	$3(3) - 2(4) = 1$	per finger
Two symmetric fingers	$2\times 1 = 2$	structural
Tendon sym. constraint (θ_R=θ_L)	-1	constraint
Net finger closure DOF	**1**	**Actuated**
Passive (spring-loaded):		
Wrist revolute ($k_w = 5.0$ N·m/rad)	$+1$	passive
Right distal joint ($k_d = 0.6$ N·m/rad)	$+1$	passive
Left distal joint ($k_d = 0.6$ N·m/rad)	$+1$	passive
Total DOF	**4**	
Actuators	**1**	
Underactuation degree	**3**	

The three passive DOFs serve **heterogeneous** functions with distinct activation conditions:

1. **Wrist** (φ): activates when actuator force exceeds $k_w \cdot \varphi_{\text{min}}$—i.e. *after* finger joints stall on a grasped object or reach their joint limit (Phase 2).
2. **Right distal** (δ_R): activates on contact between the right proximal pad and the object; deflection proportional to normal force divided by $k_d = 0.6$ N·m/rad.
3. **Left distal** (δ_L): same, mirrored about the centreline.

This distinguishes the design from conventional underactuated hands, where all passive DOFs serve the same grasping function.

Grashof condition. The longest link is $L_1 = L_3 = 40$ mm; the remaining three sum to $40 + 12 + 12 = 64$ mm. Since $40 < 64$, the Grashof condition holds and the linkage is a Class-I crank-rocker—no lock-up within $[0,\ 0.8]$ rad.

4. Forward Kinematics

Reference frame. Origin at the base centroid, x rightward, z downward (gravity).

4.1 Four-Bar Forward Kinematics

Driver pivot A sits at $(+24,\ 0)$ mm; follower pivot B at $(+36,\ 0)$ mm. For driver angle θ_1 (positive = inward closure):

$$C = A + L_1 \begin{pmatrix} -\sin\theta_1 \\ -\cos\theta_1 \end{pmatrix}, \tag{4}$$

$$D = B + L_3 \begin{pmatrix} -\sin\theta_3 \\ -\cos\theta_3 \end{pmatrix}, \quad \theta_3 = \theta_1. \tag{5}$$

The parallelogram property guarantees $|CD| = L_2 = 12$ mm for all θ_1; closure residual is numerically zero at $\theta_1 \in \{0°,\ 15°,\ 30°,\ 46°\}$. The left finger mirrors the right: $A_L = (-24,\ 0)$, $B_L = (-36,\ 0)$, with $\sin\theta_1 \to -\sin\theta_1$.

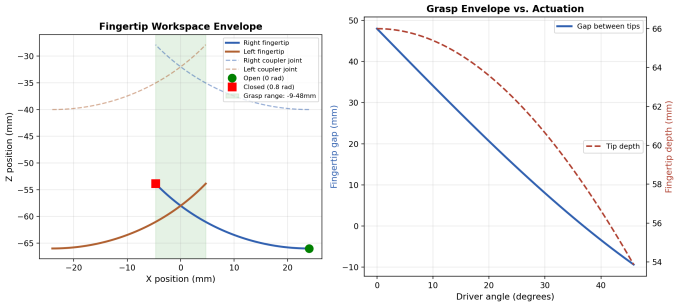

Figure 2: Fingertip workspace and grasp aperture. *Left*: right (blue) and left (orange) fingertip loci over $\theta_1 \in [0,\ 0.8]$ rad; green shading marks the graspable diameter band. *Right*: inter-fingertip gap (solid) and tip depth (dashed) vs. driver angle.

4.2 Distal Fingertip Position

The distal phalanx ($L_d = 26$ mm) hangs below C and deflects by δ when in contact:

$$P_{\text{tip}} = C + L_d \begin{pmatrix} -\sin(\theta_1 + \delta) \\ -\cos(\theta_1 + \delta) \end{pmatrix}. \tag{6}$$

At rest $\delta = 0$; under contact, the joint deflects until $k_d\,\delta$ balances the contact moment.

4.3 Wrist Angle from Tendon Constraint

Applying symmetry ($\theta_R = \theta_L = \theta_f$) to Eq. (3):

$$\varphi = 4\big(\ell_{\text{ctrl}} - 2\,\theta_f\big). \tag{7}$$

In Phase 1, fingers move freely and $\varphi \approx 0$. In Phase 2, fingers stall at $\theta_f = 0.8$ rad:

$$\varphi = 4\,(\ell_{\text{ctrl}} - 1.6)\,. \tag{8}$$

At $\ell_{\text{ctrl}} = 2.0$, $\varphi = 1.6$ rad (91.7°); at $\ell_{\text{ctrl}} = 2.5$, $\varphi = 3.6$ rad (206°) in the rigid limit—spring compliance causes a small deviation, characterised in Section 6.

4.4 Force Transmission

Actuator torque maps to fingertip normal force as:

$$F_{\text{tip}} = \frac{\tau_{\text{act}}}{L_1 \cos(\theta_1 + \delta)}. \tag{9}$$

The singularity at $\theta_1 + \delta = 90°$ is never reached: the 46° driver limit and typical contact angles keep the denominator above $\cos(46°) \approx 0.69$, bounding the transmission ratio throughout normal operation.

5. Workspace, Singularities, and Operational Issues

Workspace. At $\theta_1 = 0$ the pad faces sit at ± 16 mm from the centreline (32 mm open gap); at $\theta_1 = 0.8$ rad they converge inward. Including distal phalanx extension, the graspable diameter range is **8–36 mm** (confirmed by the diameter sweep in Section 6.4).

Singularities. The four-bar singularity (driver–coupler–ground collinear) occurs at $\theta_1 = 90°$, well outside the 46° joint limit and unreachable in normal operation. The wrist is a pure revolute with no singularity over $\varphi \in [-2\pi,\ 2\pi]$.

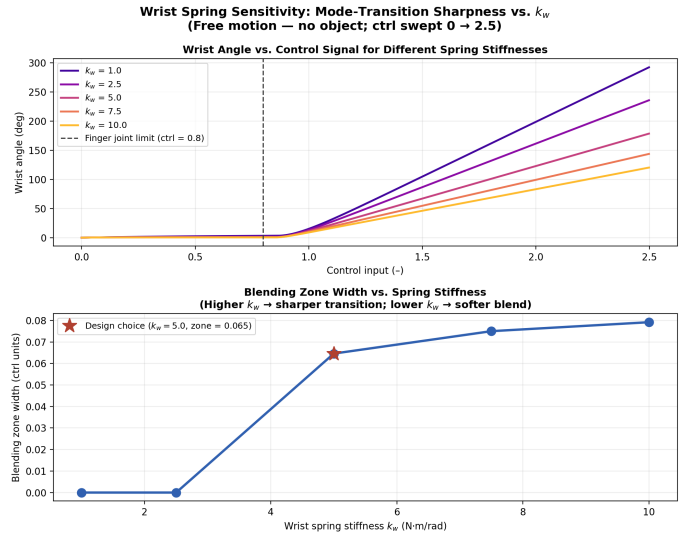

Figure 3: Wrist spring sensitivity. *Top*: wrist angle vs. control input for $k_w \in \{1.0,\ 2.5,\ 5.0,\ 7.5,\ 10.0\}$ N·m/rad (free motion, no object). *Bottom*: blending-zone width vs. k_w; red star marks the design choice $k_w = 5.0$, which balances switch sharpness against impulse loading ($\Delta\ell = 0.065$).

Mode-transition blending zone. The linear wrist spring means the Phase 1 → Phase 2 switch is not perfectly sharp: simulation ($k_w = 5.0$ N·m/rad, Fig. 3) shows a blending zone of $\Delta\ell \approx 0.065$ ctrl units where both subsystems move simultaneously. This is mechanically beneficial—it prevents impulsive joint loads at the mode boundary. Higher k_w narrows the zone (sharper switch); lower k_w widens it, as shown in Fig. 3.

6. MuJoCo Simulation

6.1 Simulation Setup

All simulations use **MuJoCo 3.3.3** with the gripper defined in `grasp_twist_gripper.xml`. Settings target stable stiff-contact and closed-chain behaviour: $\Delta t = 1$ ms, implicit integration, Newton solver, position actuator ($k_p = 80$, ± 150 N). The four-bar loop uses two `<connect>` constraints; the tendon is a `<fixed>` element with gear ratios 1.0, 1.0, 0.25.

6.2 Experiment 1: Phase Transition

Fig. 4 is the central result. ℓ_{ctrl} ramps $0 \to 0.8$ over 2 s (Phase 1), then $0.8 \to 2.2$ over 3 s (Phase 2), held 1 s, reversed. Object: cylinder, radius 10 mm.

Phase 1: Fingers close; contact stalls the right driver at $\approx$0.47 rad with $\approx$19.2° of wrist compliance already present—a *blended* handoff, not a clean switch.

Phase 2: Wrist rotates from 19.2° to 2.62 rad (149.9°) at ctrl = 2.2. Eq. (8) predicts 2.4 rad for ideal rigid-finger stall at 0.8 rad; the excess reflects earlier contact-induced stall leaving more tendon for the wrist.

Phase portrait: Strongly anisotropic, with a rounded corner consistent with compliant blending—*approximately sequential*, not an ideal L.

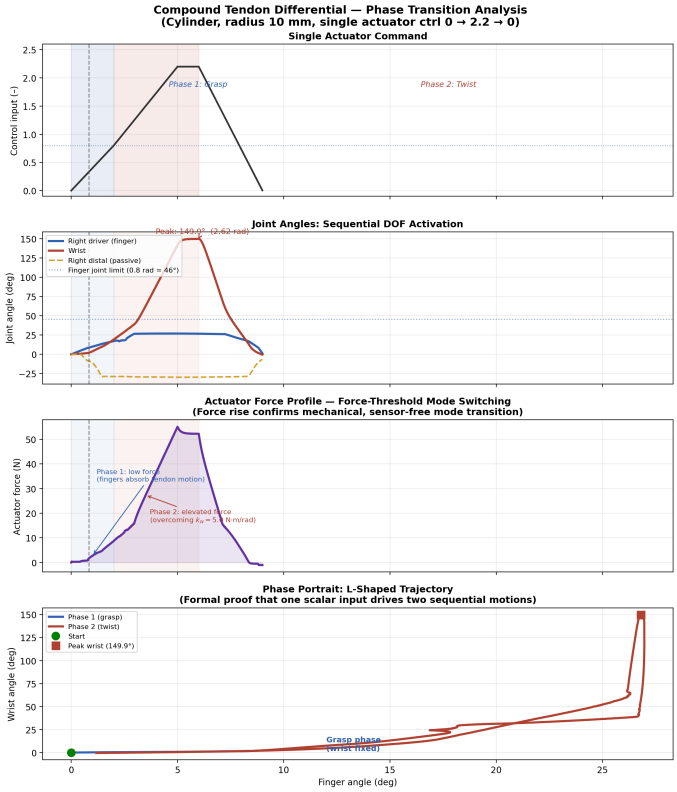

Figure 4: Phase-transition analysis, cylinder (radius 10 mm). *P1*: control signal; blue = Phase 1, red = Phase 2. *P2*: joint angles — finger stalls at ≈0.47 rad on contact; wrist dominates thereafter. *P3*: actuator force spike at finger stall. *P4*: bent phase portrait (finger vs. wrist) — approximately sequential non-coplanar motion from one input, no sensors.

6.3 Experiment 2: Multi-Shape Grasping

Cylinder (r 10 mm, h 16 mm), sphere (r 12 mm), and box ($10 \times 10 \times 16$ mm) were tested identically (all 20 g, $\mu = 1.2$; ctrl $0 \rightarrow 0.8$ in 1.5 s, $0.8 \rightarrow 2.2$ in 3 s, reversed). Peak distal deflections—cylinder 29.9°, sphere 31.6°, box 32.7°—follow contact geometry: convex surfaces produce smaller moment arms than flat faces. All three objects were retained through Phase 2, confirming geometry-independent switching.

6.4 Experiment 3: Diameter Sweep

Cylinders swept 8–36 mm (4 mm steps, ctrl $0 \rightarrow 0.8$ in 3 s; success: Z-drop < 5 mm). Reliable grasping spans **8–36 mm**; outside this range the pads either cannot generate sufficient force or cannot close far enough. *Distal deflection scales inversely with diameter—smaller objects wrap more.*

6.5 Simulation Screenshots

Fig. 7 confirms all three behaviours visually. The red arrow on the wrist disc indicates rotation; its displacement in panel (d) shows the wrist angle produced by Phase 2.

7. Conclusions

A 1-DoF underactuated gripper extracts three passive mechanical functions—wrist rotation, right and left distal conformation—from one position-controlled actuator, with no sensors, no switching logic, and no hardware beyond a compound tendon and two

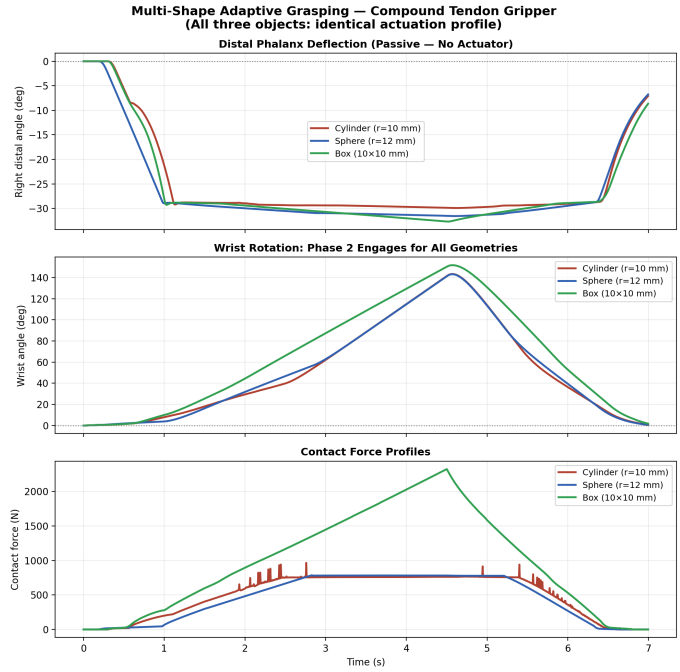

Figure 5: Multi-shape results (cylinder, sphere, box). *Top*: δ_R — geometry-dependent passive adaptation. *Middle*: wrist angle, confirming geometry-independent mode switch. *Bottom*: contact force, confirming grasp retention.

springs. The mechanism self-selects which subsystem moves based on instantaneous load: the wrist coefficient (0.25) and spring stiffness ($k_w = 5.0$ N·m/rad) bias the motion toward finger closure first, followed by wrist-dominated rotation after contact-limited stall. In practice the handoff is blended rather than perfectly sharp, but no control intelligence is required.

MuJoCo 3.3.3 simulation confirmed: (1) approximately sequential non-coplanar actuation from a monotone input (Fig. 4); (2) geometry-adaptive distal deflection across cylinder, sphere, and box (Fig. 5); (3) reliable grasping over **8–36 mm** diameter (Fig. 6).

The approach extends Nishimura *et al.* [6] and the SPINE Gripper [7] with a simpler, fully tendon-based implementation expressible directly in MuJoCo's `<fixed>` element.

Future work. The current routing allows only unidirectional wrist rotation; a counter-wound antagonistic tendon or a three-finger topology could enable bidirectional twist. A principled relationship between k_w and blending-zone width would also guide future design iterations.

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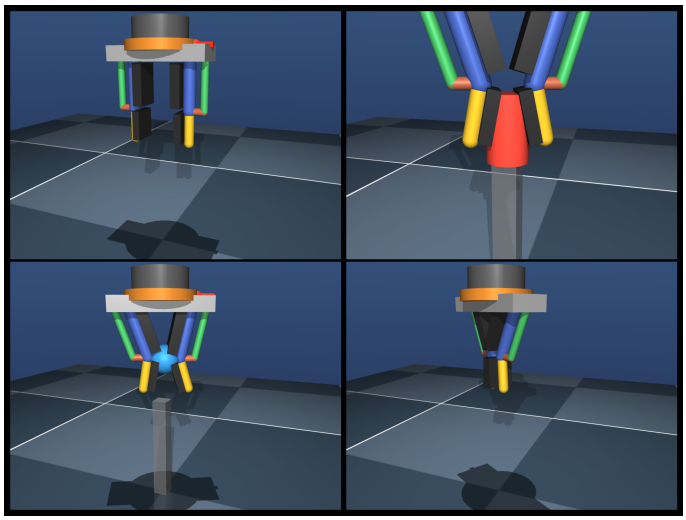

Figure 7: MuJoCo screenshots: (*a*) open (ctrl $= 0$); (*b*) cylinder mid-closure, drivers contact-limited, distals conforming; (*c*) sphere at ctrl $=$ 0.8; (*d*) free-space Phase 2 (ctrl $= 2.2$), wrist fully rotated.

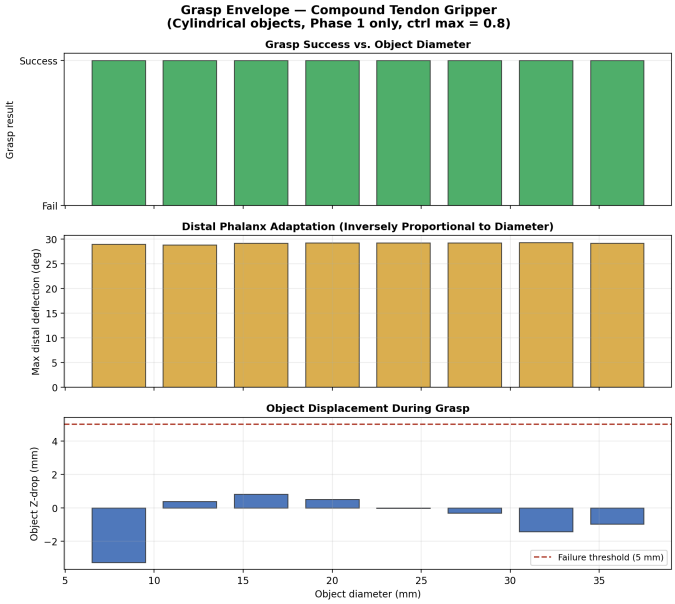

Figure 6: Grasp envelope, cylinders 8–36 mm (4 mm steps, Phase 1 only). *Top*: green = success (Z-drop < 5 mm), red = failure. *Middle*: peak distal deflection. *Bottom*: Z-displacement; dashed = 5 mm threshold.

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